

Unit -3 Logic Family

Logic Family

Saturated

- * 1. RTL
(Register-Transistor logic)
- * 2. DTL
(Diode-Transistor logic)
- * 3. TTL
(Transistor-Transistor logic)
- 4. HTL
(High threshold logic)
- 5. I²L
(Integrated Injection logic)

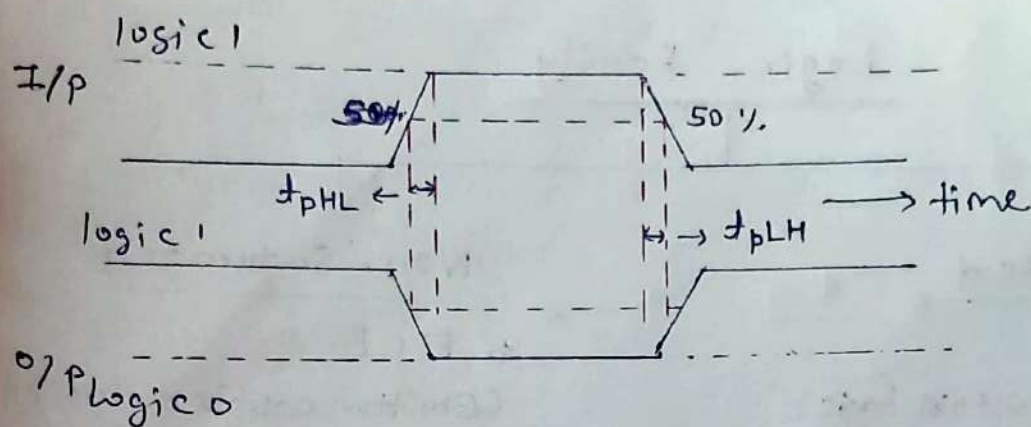
Non-Saturated

- * 1. ECL
(Emitter coupled logic)
- 2. Short key TTL
(Transistor - Transistor Logic)

* Characteristics of Digital IC

- 1. Speed of operation.
- 2. Fan in
- 3. Fan Out
- 4. Power dissipation
- 5. Noise immunity
- 6. Operating temperature.
- 7. Power supply requirement.

(1) Speed of operation



The speed of operation of an IC is expressed in terms of propagation delay. Propagation delay is defined as the time taken for the output of a gate to change after the input has changed. There are two types of propagation delay.

(1) t_{pHL} It is the propagation delay time in going from logical High to logical low.

(2) t_{pLH} It is the propagation delay time in going from logical low to logical high.

The delay times are measured between the 50% voltage level of the input & output waveform.

The avg. of two propagation delay times

$$= \frac{t_{pHL} + t_{pLH}}{2}$$

* ⁽²⁾ Power Dissipation

Power dissipation is measure of the power consume of logic gate. fully driven by all the input.

It is expressed in miliWatt (mW) & nanoWatt (nW)

* ⁽³⁾ Fan-in

The Fan-in of a gate is the number of input connected to the gate without any degradation in the voltage level. An 8 input-gate required 1 unit load per input, its Fan-in is 8.

* ⁽⁴⁾ Fan-out

The Fan-out is the maximum number of similar logic gate that a gate can drive without any degradation in the voltage level.

* ⁽⁵⁾ Noise Immunity or Noise Margin

The Noise denote an "unwanted signal". Noise can sometimes cause the input voltage of a logic gate to drop below (minimum) or rise above which leads to unreliable operations. Noise immunity is the maximum noise voltage that will may appear at the input of logic gate without changing the logical state of its output. The quantity measure by Noise immunity is called Noise margin.

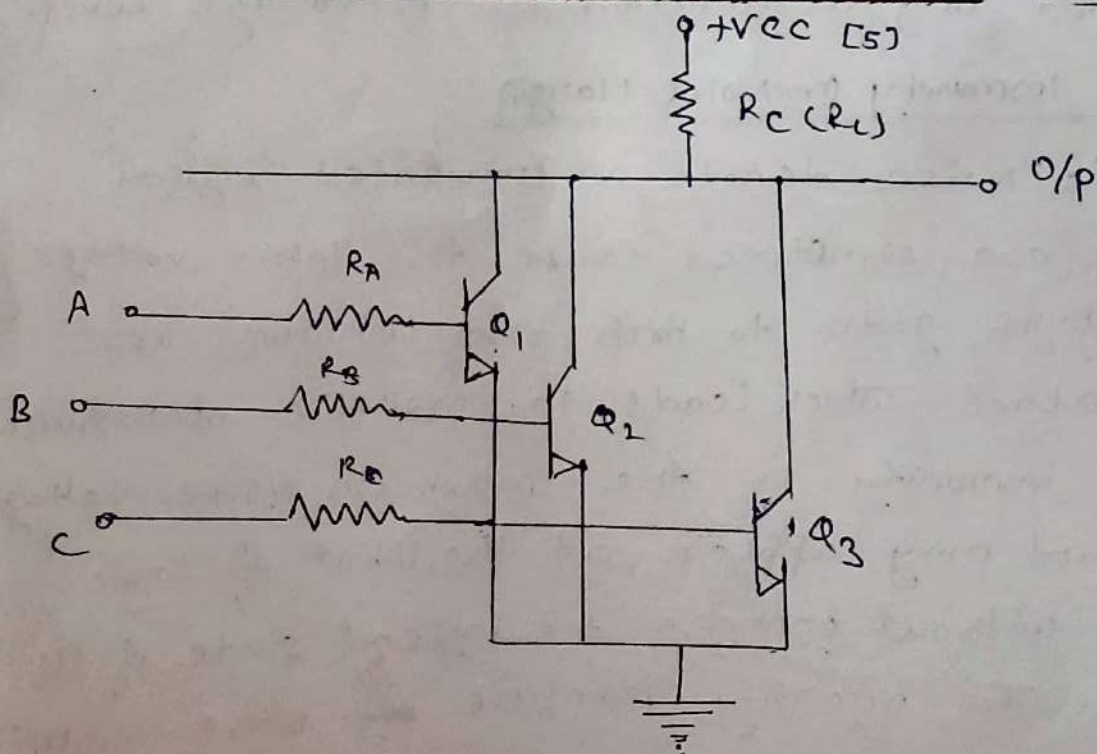
(6) * Operating Temperature

All IC are semiconductor devices that are temperature sensitive by nature. The operating temperature range for an IC vary from 0°C to 70°C for consumer and Industrial purpose. And from -55°C to $+125^{\circ}\text{C}$ for military purpose.

(7) * Power Supply Requirement

The amount of Power supply voltage required by an IC are the main parameter to be taken a proper input/output power supply.

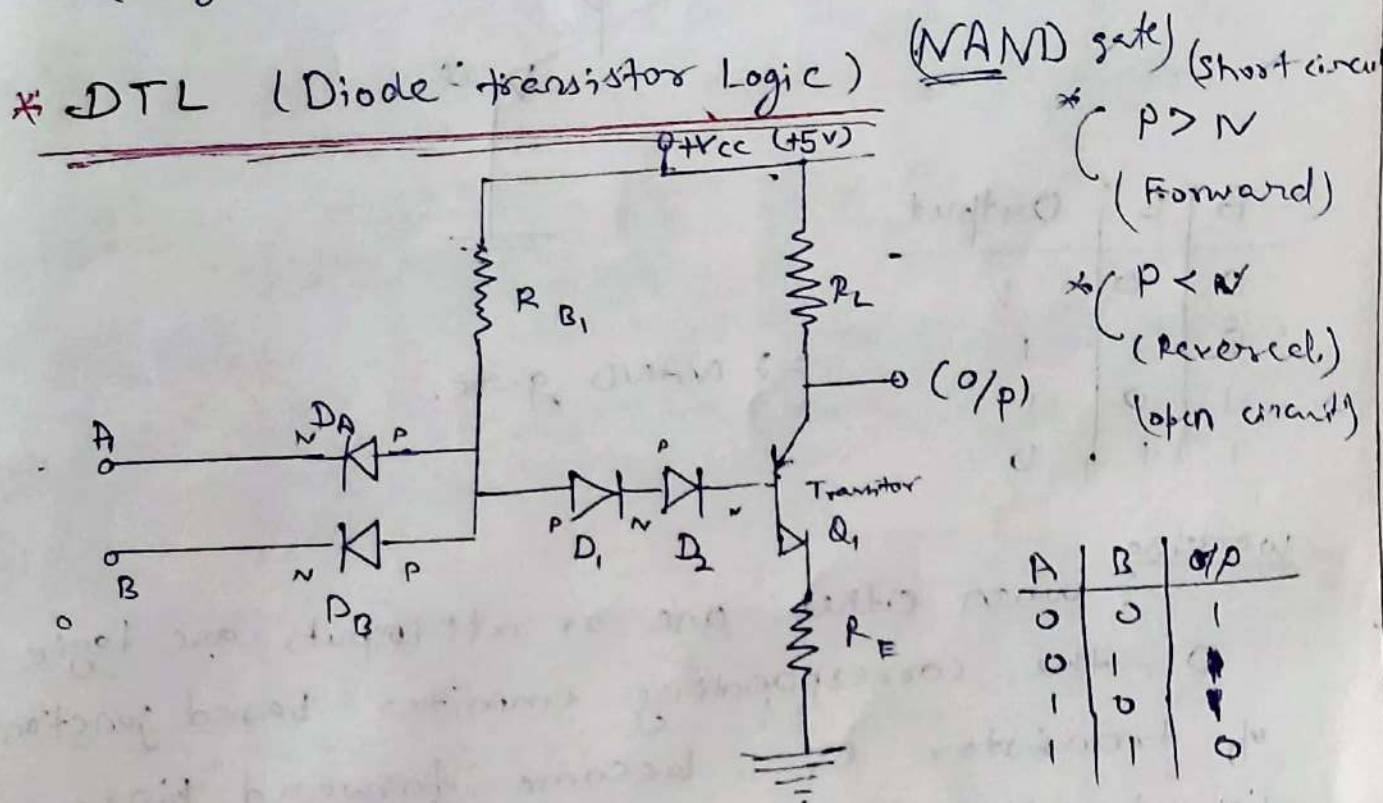
* RTL (Register Transistor Logic) (NOR gate)



Working

The RTL (NOR gate) consisting of resistors & transistors when all the input A, B & C are zero volt. All transistor are turn off,

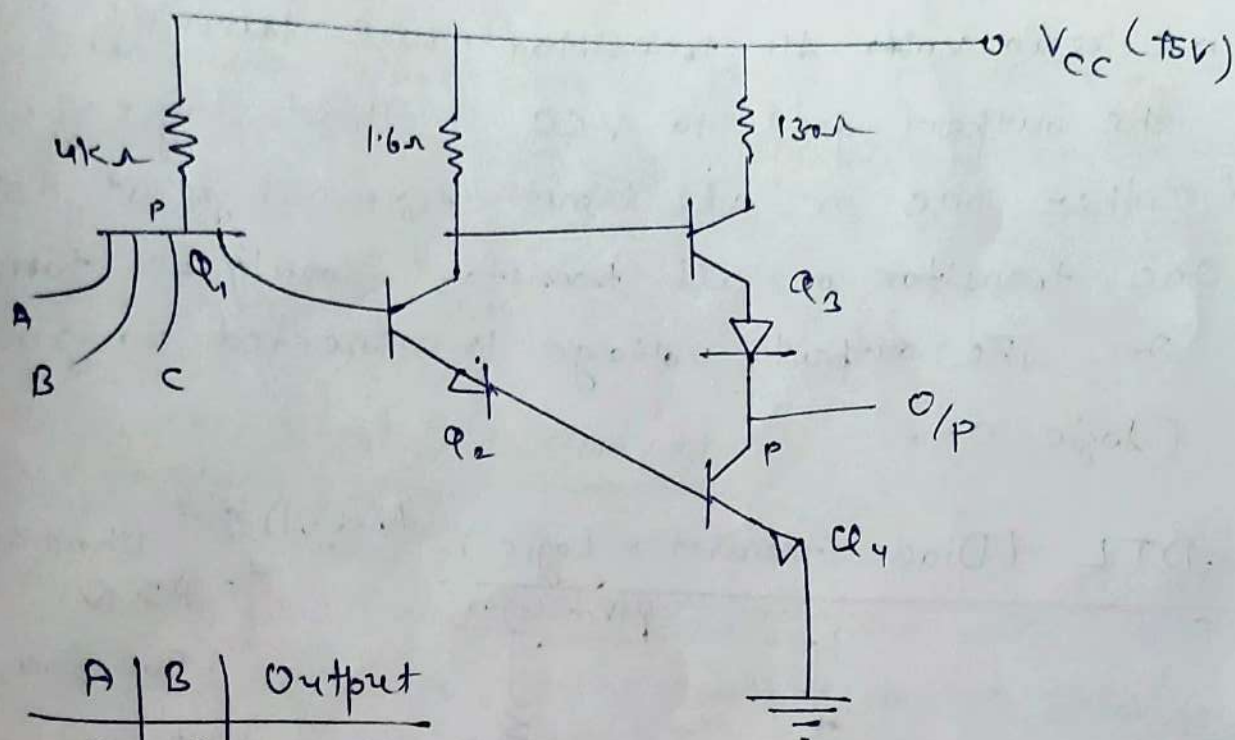
the output goes to VCC is that logic 1. if either one or all input terminal at +5V one transistor or all transistors would be turn on. The output voltage is connected to ground (logic 0).



Working

The DTL (NAND gate) consisting of resistors & transistors when all the input A & B are zero volt. All the diode DA & DB are turn on then all voltage will drop out & voltage of R_L will come output that is logic 1.

* TTL (Transistor - Transistor Logic) ^{*(Imp)}



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

⇒ NAND gate

Working

When either one or all inputs are logic 0, the corresponding emitter-base junction of transistor Q_1 becomes forward biased & transistor Q_1 is turn-on.

The value of i_{b2} is reduce so that transistor Q_2 & Q_4 are turned off. The value of base current i_{b3} is high. The transistor Q_3 is turn on. The diode p terminal is connected to +ve voltage & n terminal to 0 voltage.

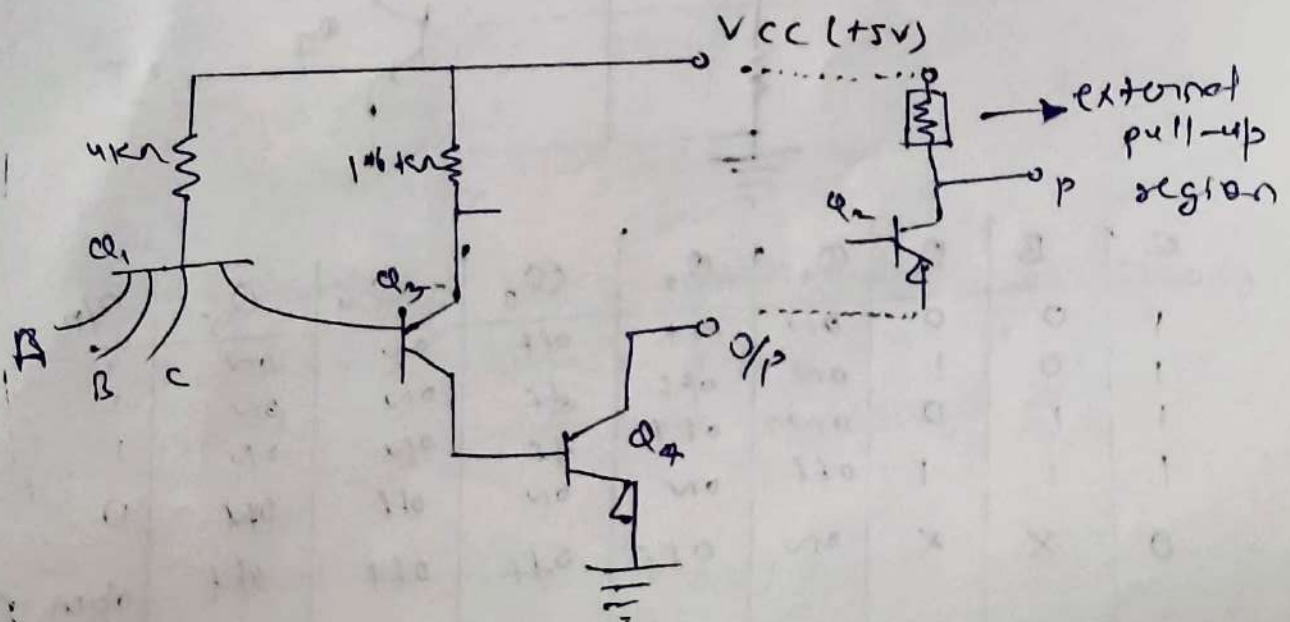
Diode become in forward bias. Output is directly connected to V_{CC} means logic 1.

When all input are high the emitter base junction of Q_1 transistor is reverse biased, the collector base junction of Q_2 transistor is forward biased. This means large current will apply on Q_2 & Q_3 is turn on. The emitter terminal is directly connected to Q_4 . So Q_4 is turned on. Output is directly connected to ground (0).

TTL output Parameter

- (i) Totempole output
(working & diagram same as TTL NAND)
- (ii) Open collector output
- (iii) Tri-state output.

(ii) Open collector output :-



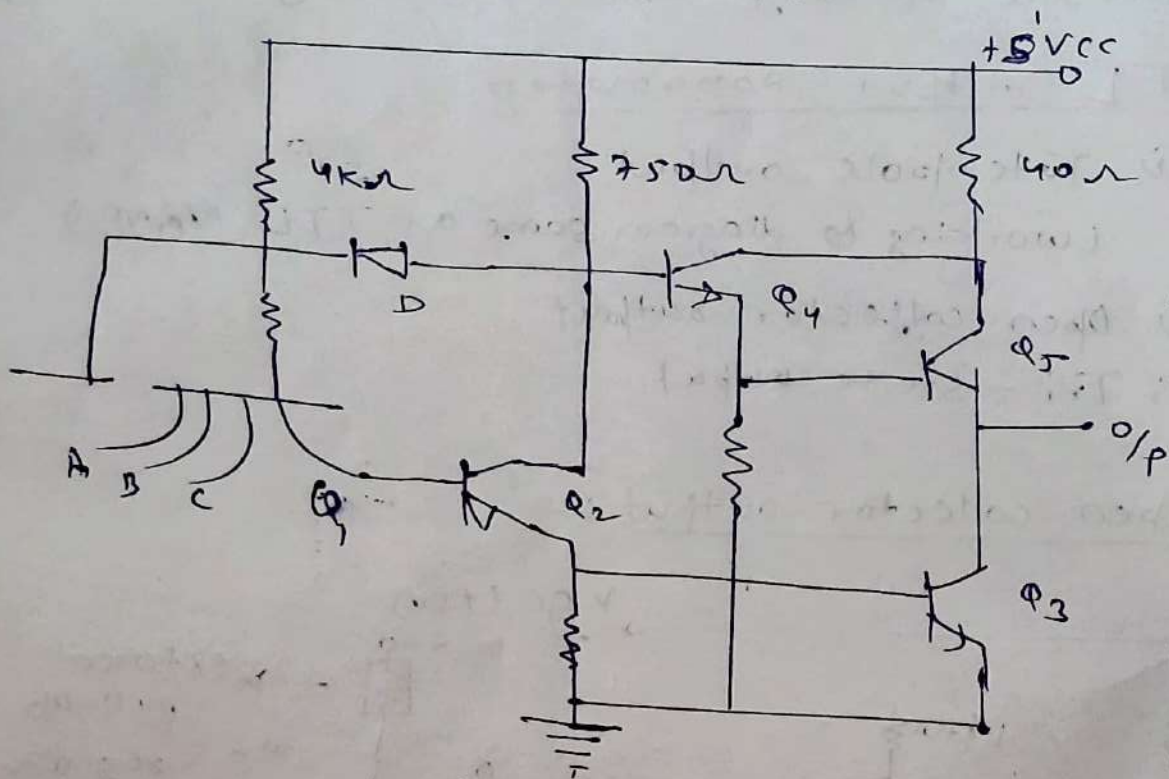
Working

The TTL NAND gate with open collector output is obtained by removing the following component. Transistor Q_3 diode Q_4 & R_3 resistor. The collector of Q_4 is open. This open collector gate will not work properly.

The circuit diagram of the TTL NAND gate with open collector output is shown below.

The open collector TTL need an external pull-up resistor to connected between transistor & power supply.

(iii) Tot - state output :-



C	B	A	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	O/p
1	0	0	ON	OFF	OFF	ON	ON	1
1	0	1	ON	OFF	OFF	ON	ON	1
1	1	0	ON	OFF	OFF	ON	ON	1
1	1	1	OFF	ON	ON	OFF	OFF	0
0	X	X	ON	OFF	OFF	OFF	OFF	open circuit

* MOSFET (Metal oxide Semiconductor field effect transistor)

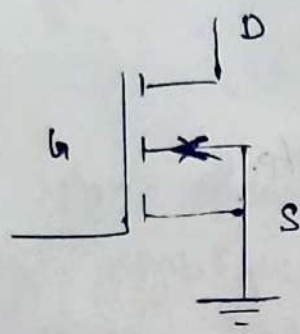
A MOSFET is a voltage-controlled device where current flows between Drain & source controlled by gate voltage.

Terminals -

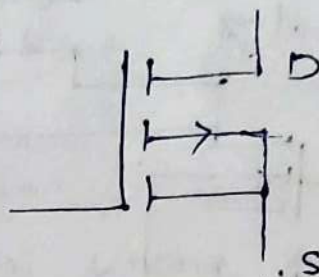
Gate (G) - Control terminal

Drain (D) - current entry

Source (S) - current leaves.



N-channel.



P-channel

→ N-channel works only when gate voltage is (+)ve.

→ P-channel works only when gate voltage is (-)ve.

Types

(i) NMOS →

(ii) PMOS -

PMOS (mainly logic family works on PMOS)

0 → ON → hole uses (-) supply.

1 → OFF. low hole mobility.

NMOS

gate low → No channel

gate high → electrons from channel → ON ✓

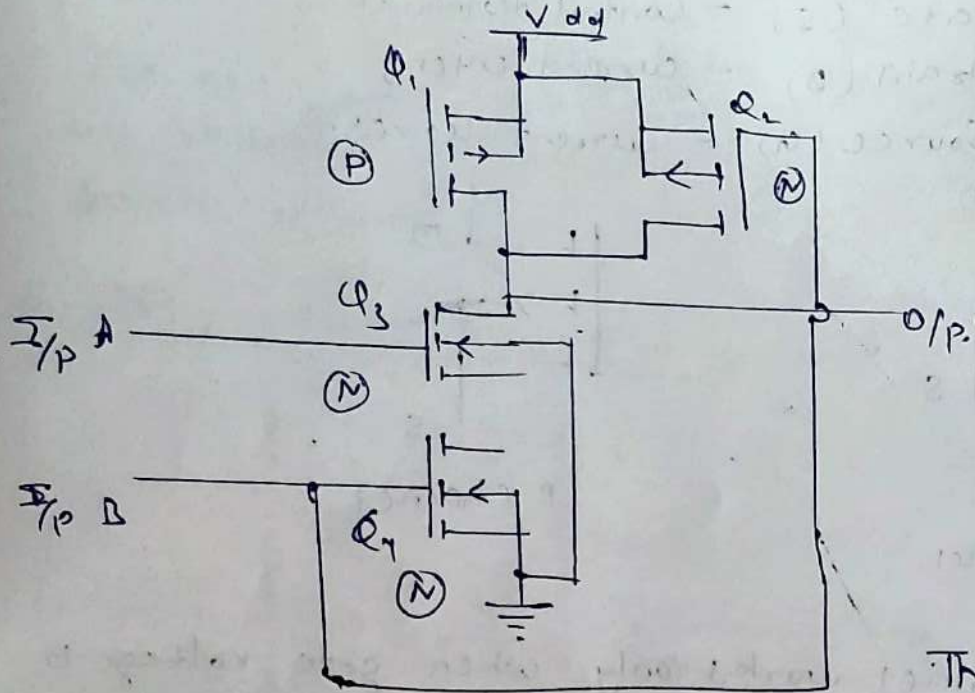
Combination of NMOS + PMOS = CMOS

*CMOS

*CMOS inverter →

PMOS → Pull up Network

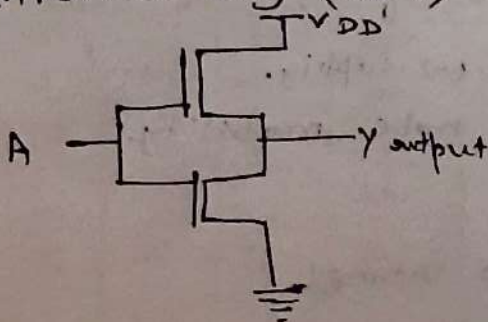
NMOS → Pull down Network



This is circuit of
CMOS NAND

A	B	Pull-up	Pull Down	Y
0	0	ON	OFF	1
0	1	ON	OFF	1
1	0	ON	OFF	1
1	1	OFF	ON	0

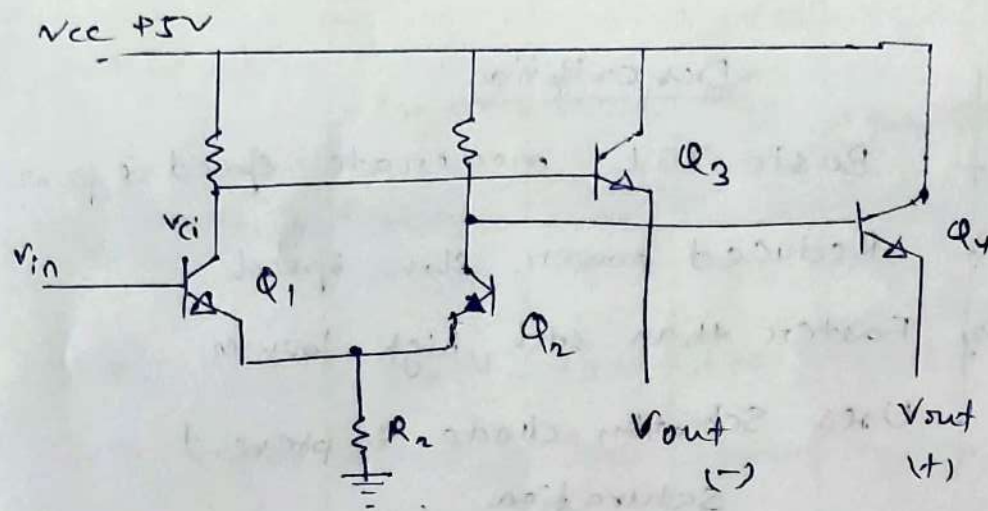
* for Inverter only (NOT)



A	Y
0	1
1	0

(NOT)

* ECL (Emitter Coupled Logic) (NOR gate)



High speed logic family. uses BJT differential amplifier circuits. Where transistor never enter saturation.

logic level are obtained by current steering not voltage switching.

working

Input voltage decides which transistor conducts. One transistor ON - other off.

No saturation.

output becomes High only when all inputs are low.

A	B	output
0	0	1
0	1	0
1	0	0
1	1	0

$\Rightarrow$ NOR gate

* TTL Subfamilies

Developed to improve speed, power or both

<u>Subfamily</u>	<u>Description</u>
1. Std. TTL	Basic TTL, moderate speed & power
2. L (Low Power)	Reduced power, slow speed
3. H (High Power)	Faster than std, high power
4. S (Schottky)	Uses Schottky diode to prevent saturation.
5. LS (Low power Schottky)	Low power + good speed. (Commonly used)
6. AS (Adv. Schottky)	Very high speed (fastest)
7. ALS (Adv. low power Schottky)	Lower power + improved speed. (best)

* Speed

$AS > S > LS > std. > L$

* Power

$H > S > AS > LS > ALS > L$

* Comparison of Logic family

parameter	RTL	DTL	TTL	ECL	MOS	CMOS
Active device	BJT + Resistor	BJT + diode	BJT	BJT (differential)	MOSFET	NMOS + PMOS
Logic operation	Resistor logic	Diode logic	Transistor logic	Current Steering	Voltage Controlled	Complementary
Supply V.	+5V	+5V	+5V	-5-2V	5-15V	3-15V
Switching speed	Very low	low	Medium	Very High	Medium	High
Power dissipation	High	High	Medium	Very High	Medium	Very low
Noise Margin	Poor	Better (RTL)	Good	Low	Good	Best
Fan-out	Very low	low	High	low	High	Very High
propagation delay	High	high	Medium	Very low	Medium	low.